DESCENT DATA FOR CLOSED IMMERSIONS OVER AN FPQC COVERING ARE EFFECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A closed immersion is affine. Morphisms, Lemma morphisms-lemma-closed-immersion-affine, invoked at proof lines 9435-9436. Verifies the hypothesis of the affine effectivity lemma.
- Descent data for affine morphisms over an fpqc covering are effective. Same chapter, Lemma lemma-affine, invoked at proof lines 9436-9437. Gives effectivity of the datum.

Source reference. Original lemma 0245: sources/stacks-descent/pinned-original-source.tex, source lines 9365–9365. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01SE: sources/stacks-chow/tags.txt, source lines 2335–2335. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 022B: sources/stacks-chow/tags.txt, source lines 2700–2700. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\{X_i \to S\}_{i \in I}$ be an fpqc covering, see Topologies, Definition 022B. Let $(V_i/X_i, \varphi_{ij})$ be a descent datum relative to $\{X_i \to S\}$. If each morphism $V_i \to X_i$ is a closed immersion, then the descent datum is effective.*

Proof. This is true because a closed immersion is an affine morphism (Morphisms, Lemma 01SE), and hence Lemma 0245 applies. □

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